

Muthui Mureithi

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SUMMARY OF SKILLS

Tools: PyCharm • Visual Studio Code • ModelSim • Xilinx Vivado • MatLab • Arduino • Power BI

Programming: C++ • Python • Java • VHDL • Javascript

Languages: French • English • Japanese

Operating system: Windows 11, Linux

EDUCATION

Bachelor of Engineering (Computer Engineering)

2022 - 2028

Concordia University, Montreal, QC

PROJECTS

QualiAir Hardware and Application System (Academic, Group)

2026

- Tools and languages: C++, Java, Python, Android Studio, ESP32
- Designed and built an installation equipped with gas sensors, a dust sensor, and a temperature sensor to determine air quality.
- Designed and built an Android application capable of receiving notifications and displaying surrounding air quality data from the installation to the user.

AI model for Sentiment Analysis (Academic, Group)

2026

- Tools and languages: Python
- Designed and built an AI model for sentiment analysis on Amazon reviews for math, science, and engineering textbooks.
- Used a Naïve Bayes model, achieving an F1 score of 0.65.

Real-time Chatting Application (Academic, Group)

2025

- Tools and languages: Node.js, React.js, MySQL
- Built a full-stack chat application where users can send/receive messages in real time.
- Implemented Agile practices (short sprints, feature testing) and Git-based team collaboration.
- Strengthened skills in REST API development, relational database design, and React state management.

Hovercraft Competition (Academic, Group)

2024

- Tools and languages: Atmega328p, C/C++, Arduino
- Developed and built a hovercraft capable of navigating a maze using ultrasonic sensors and fans.
- Programmed microcontroller logic in C/C++ and Arduino.

WORK EXPERIENCES

Intern

May 2026 - Aug 2026

Polytechnique de Montréal, Montreal, Quebec

- Upgraded a flight controller and validated it through a FlightGear flight simulation.
- Designed and built a data acquisition system to measure wind velocity using a seven-hole pitot tube and an Arduino Due. The collected data was used to train an MLP model to predict the wind velocity vectors. The resulting model achieved an error below 9%.

VOLUNTEER WORKS

Volunteer at HackDécouverte

Nov 2023

Bilingual Hackathon for cégep students, Montreal, Quebec

Volunteer at Otakuthon

Aug 2023

Convention promoting Japanese animation, Montreal, Quebec

INTERESTS

- Badminton
- Video games
- Violin